Progress B Supervisor Meeting Preparation

Credit Card Fraud Detection using a Hybrid Autoencoder-XGBoost Model with Local LLM Explanations

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Private study and consultation guide - *not part of the academic report*

How to use this guide

- Part I contains the meeting story, exact figures, technical explanations and live-demo order.
- Part II contains sixty likely supervisor questions with concise answers.
- Use the short answer first. Add technical detail only when the supervisor asks a follow-up question.
- The academic report remains the source to give the supervisor; this guide is private preparation material.

Part I - Meeting briefing

Student: NG YI ZHEN **Student ID:** 23076003 **Project:** Evidence-Grounded Local-LLM Explanations for Credit Card Fraud Alert Review **Meeting preparation date:** 29 July 2026

This is a private preparation document. Do not insert it into the academic report.

1. What to bring

Bring or open these files:

1. The clean Progress B PDF:

`CP2/02_SUPERVISOR_WORKSPACE/01_SEND_TO_SUPERVISOR/Meeting_01_Progress_B/CP2_Working_Draft_Chapters_1_to_4.pdf`

2. The running local application at `http://127.0.0.1:8000/queue`.
3. This quick preparation file.
4. `01_COMPLETE_PROJECT_EXPLANATION_BILINGUAL.md` if the supervisor asks for deeper technical detail.
5. `02_SUPERVISOR_QUESTION_BANK.md` for likely follow-up questions.

The final CP2 report and presentation now contain the correct detector seeds: **42, 43, 44, 45, and 46**.

2. Your project in one sentence

这个项目建立并评估一个可审计的信用卡诈骗调查流程：冻结的 XGBoost detector 负责筛选交易，SHAP 负责提供模型贡献证据，本地 LLM 只负责把证据写成候选叙述，确定性 guardrail 决定叙述能否交付，失败时系统退回 reason codes，最后由人类分析师处理案件。

> ***English answer*** *This project implements and evaluates an auditable fraud-review pipeline in which a frozen detector prioritises cases, SHAP produces signed model evidence, a local LLM proposes a short narrative, deterministic guardrails control whether that narrative may be delivered, and a human analyst retains responsibility for the case decision.*

3. Sixty-second explanation

> *Credit-card fraud is a highly imbalanced classification problem, so accuracy alone is misleading. I first compared six detector configurations across five fixed seeds using leakage-controlled preprocessing and Average Precision as the primary metric. The configurations included direct XGBoost baselines, SMOTE, cost-sensitive learning, autoencoder reconstruction error and latent features. The autoencoder variants did not show a clear detector-level advantage, so the analyst workbench uses the frozen G6 seed-42 cost-sensitive XGBoost detector. SHAP then generates directional reason codes for the 51 flagged test cases. A local Llama 3 8B model may translate those reason codes into short narratives, but the text is treated as untrusted. Four deterministic checks verify format, completeness, grounding and direction. Failed or unavailable narratives are replaced with deterministic reason-code fallback. The React and FastAPI workbench demonstrates the complete analyst workflow while keeping research evidence read-only and analyst notes in a separate SQLite workflow store.*

4. The five responsibility boundaries

Memorise this sequence:

> ***The detector decides what to flag. SHAP explains the detector output. The LLM proposes wording. The validator controls delivery. The analyst makes the provisional decision.***

Never say the LLM detects fraud. Never say SHAP proves the real-world cause of fraud.

5. Numbers you must know

Dataset

Item	Number
Original transactions	284,807
Original fraud cases	492
Original fraud prevalence	approximately 0.1727%
Transactions after content deduplication	283,726
Fraud cases after deduplication	473
Seed-42 train split	198,608 transactions, 331 frauds
Seed-42 validation split	42,559 transactions, 71 frauds
Seed-42 test split	42,559 transactions, 71 frauds

Experiment design

Item	Value
Detector groups	G0, G1, G2, G3, G6 and G7
Fixed seeds	42, 43, 44, 45 and 46
Detector runs	6 groups x 5 seeds = 30 runs
Frozen workbench detector	G6 seed 42
G4	SHAP reason-code stage
G5	Local-LLM narrative stage

Frozen detector

Metric	Result
Validation AP	0.877447
Test AP	0.820176
Test ROC-AUC	0.977399
Threshold	0.989038
Precision	0.980392
Recall	0.704225
F1	0.819672
True positives	50
False positives	1
False negatives	21
True negatives	42,487
Flagged cases	51

Plain-language interpretation:

> *The frozen detector created a very precise review queue: 50 of the 51 flagged cases were fraud in the test data. However, it did not identify all fraud and missed 21 of the 71 fraud cases.*

Guardrail experiment

Result	Value
Versioned adversarial samples intercepted	330/330
Template-compatible faithful controls accepted	318/318
Strict raw outputs with detected violation	2/51, 3.92%
Strict narratives delivered	49/51
Strict fallbacks	2/51
Simple raw outputs with detected violation	51/51, 100%
Simple fallbacks	51/51
Student-conducted blinded manual semantic audit	0/49 semantic violations, 95% CI 0% to 7.27%

Correct qualification:

> *These are detected violations under the implemented validator and the evaluated prompt, model and case set. The manual audit gives bounded supporting evidence for the 49 reviewed delivered strict narratives, but it is a single-reviewer result and does not prove universal semantic correctness.*

Fresh software verification on 20 July 2026

Check	Result
FastAPI/backend dashboard tests	37 passed
React component tests	5 passed
Playwright end-to-end tests	10 passed
ESLint	passed
TypeScript/Vite production build	passed

6. Experimental groups in simple language

Group	What it uses	Why it exists
G0	Original 30 features and XGBoost	Direct baseline
G1	Original features, training-only SMOTE and XGBoost	Test data-level oversampling
G2	Original features plus AE reconstruction error	Test anomaly signal
G3	G2 representation plus training-only SMOTE	Test AE signal with oversampling
G6	Original features and cost-sensitive XGBoost	Test algorithm-level imbalance handling
G7	Original features plus 10 AE latent features	Test learned representation
G4	Frozen G6 detector and SHAP	Produce explanations, not a detector
G5	G4 evidence and local LLM	Evaluate narrative delivery, not a detector

7. The honest result

The autoencoder was properly implemented and evaluated, but did not clearly improve detection.

> ***English answer*** *Autoencoder-derived features did not demonstrate a clear and consistent detector-level advantage over the original-feature baselines on this dataset. This is a valid negative or near-null result, and it supports using the simpler frozen G6 detector in the workbench.*

Do not say G6 is statistically superior. It had the numerically highest mean AP, but the leading differences were small and no significance test was logged.

8. What is innovative

Do not claim that XGBoost, Autoencoder, SMOTE, SHAP or LLMs are individually new.

> ***Recommended English answer*** *Within the reviewed literature, the contribution is an evaluated and auditable narrative-delivery boundary for fraud detection. The project connects frozen SHAP evidence, data-minimised local generation, deterministic four-part validation, paired guardrail-off and guardrail-on measurement, quantified fallback, artifact provenance and a human analyst workflow.*

If the supervisor says this is only integration:

> *The implementation is the engineering artefact, but the research contribution is the measured question. I compare detector configurations, preserve the same raw LLM output for paired policy evaluation, calibrate a validator against adversarial and faithful controls, quantify detected failure and fallback, and document the limits of the measurement instrument.*

9. Why not just use a normalizer and XGBoost

> *A direct XGBoost pipeline is a necessary baseline and may be the better operational choice. Standardisation changes feature scale but does not create anomaly information. The autoencoder tests whether reconstruction error or latent representations add useful information beyond the original features. Because the experiment did not show a clear advantage, the workbench uses the simpler cost-sensitive XGBoost configuration. Scaling is still required for the autoencoder and distance-based SMOTE, and it must be fitted on training data only.*

Important distinction:

- A data normalizer standardises numbers.
- A deterministic narrative renderer standardises wording.
- A guardrail evaluates an unmodified LLM output and rejects it if it violates the evidence contract.

The validator must not silently repair the raw narrative before measurement, because that would hide the LLM's original failure.

10. Why keep the LLM if deterministic reason codes are safer

> *A deterministic renderer may remain the safer operational default. The LLM is optional and is not trusted as evidence. The research question is whether a requested generative narrative layer can be constrained, measured, rejected and safely replaced. This project has not yet proven that the LLM improves analyst speed, comprehension or decision quality.*

11. Why the dataset is enough and not enough

For an undergraduate FYP:

- over 283,000 deduplicated transactions;
- 473 real fraud cases;
- six detector configurations;
- five fixed seeds;
- train/validation/test separation;

- complete detector, explanation, narrative and software artifacts.

For a production bank:

- only one historical public dataset;
- anonymous PCA features;
- approximately two days of transactions;
- no customer, merchant, device or location context;
- no temporal drift study;
- no Malaysian banking data;
- no external dataset replication.

> ***English answer*** *The dataset is adequate for a controlled FYP experiment, but it is not adequate for universal industrial generalisation or a production banking claim.*

12. Data leakage answer

Answer in this exact order:

6. Split before learned preprocessing.
7. Fit the scaler on training data only.
8. Apply SMOTE to training data only.
9. Fit the autoencoder on legitimate training transactions only.
10. Use validation data for early stopping, tuning and threshold selection.
11. Freeze the model and threshold before final test evaluation.
12. Keep `case_id` for joining artifacts but exclude it from model features.
13. Use manifests and hashes to bind downstream artifacts to their source runs.

Limit you should admit:

> *The split is stratified rather than chronological. It controls class balance but does not fully simulate future temporal drift.*

13. Why Average Precision, not accuracy

If every transaction were predicted legitimate, accuracy would be about 99.83% but recall for fraud would be zero.

> *Average Precision summarises precision-recall behaviour for the minority class. Precision explains how many flagged cases are truly fraud, while recall explains how many fraud cases were detected. The operational threshold trades analyst workload against missed fraud.*

14. What SHAP means

SHAP answers which features pushed the frozen model output up or down for a case.

It does not prove:

- the real-world cause of fraud;
- the true meaning of V14 or V12;
- that a customer acted dishonestly.

> ***English answer*** *SHAP provides model-level directional attribution, not causal explanation. Because V1 to V28 are anonymised PCA components, the system must not invent business meanings for them.*

15. Guardrail logic

The four checks are:

14. **Format -** required narrative sections and no unauthorised numeric claims.

15. **Completeness -** all required reason-code evidence appears in the expected order.
16. **Grounding -** no invented or unauthorised feature or unsupported phrase.
17. **Direction -** every feature preserves the recorded SHAP direction.

Any failed check activates deterministic fallback. An Ollama outage also activates fallback.

> ***English answer*** *The validator is deterministic and closed-grammar. It provides repeatable enforcement of known evidence constraints, but it is not a universal judge of unrestricted English.*

16. What the application does

Work Queue

Shows the 51 cases flagged by the frozen detector and routes them according to analyst workflow state, explanation exceptions and detector ordering.

Investigation Workspace

Shows immutable detector evidence, signed SHAP contributions, reason codes and the recorded narrative beside a separate analyst decision rail.

Narrative Assurance

Uses the real validator to demonstrate direction flip, unlisted-feature injection and template corruption. Failure shows the rejected narrative and deterministic fallback.

Model and Policy Monitor

Separates detector performance evidence from narrative-policy evidence and exposes the recorded provenance chain.

Workflow storage

Analyst status, provisional disposition, note, activity history and revision are stored in a local SQLite database. Model scores, predictions, SHAP evidence, narratives and report artifacts are read-only. Evidence fingerprints prevent old workflow records from being treated as compatible with a different model evidence chain. Optimistic concurrency prevents a stale browser page from overwriting a newer update.

17. Recorded mode versus live mode

- **Recorded mode:** reads the exact frozen G5 narrative used in research evaluation.
- **Live mode:** asks local Ollama to regenerate wording from the same bounded evidence and then runs the real validator.
- Live mode does not retrain or change XGBoost, the threshold or SHAP evidence.
- Live output is temporary and not written back into the immutable research artifacts.
- If Ollama is unavailable, the page returns a successful fallback state rather than breaking.

18. Accurate product positioning

> *This is a functional, analyst-oriented local research prototype. It demonstrates product architecture, evidence integrity, workflow state, fallback and test coverage, but it is not a production-ready banking platform.*

Already implemented:

- React/TypeScript user interface;
- FastAPI backend;
- SQLite workflow persistence and activity history;

- immutable evidence and writable workflow separation;
- source-manifest verification and evidence fingerprints;
- optimistic concurrency;
- local-only server and Ollama host restrictions;
- recorded/live narrative separation;
- deterministic fallback;
- automated backend, component and browser tests.

Not implemented:

- real bank streaming integration;
- authentication and role-based access control;
- enterprise encryption and secrets management;
- fraud operations SLAs and escalation integration;
- model drift monitoring and retraining governance;
- regulatory certification;
- real merchant, customer, device and account-history context;
- measured analyst productivity or usability.

19. Eight-minute demo order

0:00-0:45 - State the research problem

> *The detector selects cases; SHAP provides evidence; the LLM only proposes wording; and the guardrail controls delivery.*

0:45-2:00 - Work Queue

- Explain that there are 51 flagged test cases.
- Explain that this is a review queue, not all transactions.
- Do not call the score a calibrated probability.

2:00-4:00 - Investigation Workspace

- Open case 42009 or another verified case.
- Show the fixed score and threshold.
- Show SHAP directions and reason codes.
- Show the separate analyst note/disposition area.

4:00-5:15 - Recorded narrative

- Explain the strict-prompt recorded output.
- Point to the four checks.
- Say that passing means compliance with the implemented evidence contract.

5:15-6:30 - Narrative Assurance

- Run direction flip first because it is easiest to explain.
- Show direction failure.
- Show deterministic reason-code fallback.
- Mention invented-feature and template-corruption presets.

6:30-7:15 - Model and Policy Monitor

- Show the detector groups and narrative-policy results separately.
- Open Evidence Details and explain run IDs and hashes.

7:15-8:00 - Close honestly

> *The autoencoder variants did not show a clear detector advantage. The stronger contribution is the measured and fail-safe boundary between model evidence and generated analyst prose.*

20. Questions to ask the supervisor

Ask only after presenting the current work:

18. Is the contribution positioned correctly as an evaluated narrative-delivery boundary rather than a new detector algorithm?
19. Should the final title retain Autoencoder-XGBoost because it is central to the experiment, or broaden to the complete verifiable fraud-review pipeline?
20. Is a small human blind audit necessary before submission, or may the absence of one remain an explicit limitation?
21. Which result should receive the most presentation time: detector comparison, guardrail experiment or analyst workbench?
22. Are there specific Sunway report-format or appendix expectations you want applied before final submission?

21. Statements you must never make

- The LLM detects fraud.
- SHAP proves why the transaction is truly fraudulent.
- V14 means a suspicious merchant or device.
- The model detects almost all fraud.
- A score of 0.999 is definitely a 99.9% real-world fraud probability.
- G6 is statistically superior.
- The guardrail eliminates all hallucinations.
- Zero delivered violations proves perfect semantic faithfulness.
- Local Ollama guarantees privacy, security or compliance.
- The website is production ready.
- Autoencoder definitely improves XGBoost.
- This is the first system in the world.

22. Final closing answer

> *This FYP does not claim that a new detector or an LLM can autonomously solve fraud investigation. It evaluates how established fraud detection, model attribution, local language generation, deterministic validation, provenance and human workflow can be combined without allowing the generative component to alter or silently replace the underlying evidence.*

Part II - Supervisor question bank

Use the short answer first. Add the follow-up explanation only if the supervisor continues asking.

A. Project framing

Q1. What exactly did you build?

I built a reproducible fraud-review research pipeline and a local analyst workbench. The pipeline compares detector configurations, freezes one detector, generates signed SHAP evidence, evaluates optional local-LLM narratives behind deterministic guardrails, and preserves a reason-code fallback. The application lets an analyst review the same frozen evidence and record provisional decisions separately.

Q2. Is this mainly a machine-learning project or a website project?

The research core is the evaluated detector-to-explanation pipeline. The website is the delivery artefact proving that the frozen detector, SHAP evidence, narrative policy, fallback and provenance can support a coherent analyst workflow. The website does not replace the experiments.

Q3. What changed from CP1 to CP2?

CP1 proposed the problem, literature gap, method and evaluation design. CP2 implemented and strengthened the same research objective through executable code, 30 detector runs, frozen manifests, SHAP reason codes, a local narrative experiment, adversarial validator calibration, paired guardrail evaluation and a React/FastAPI/SQLite analyst workbench.

Q4. Is CP2 a different project from CP1?

No. CP2 operationalises the CP1 proposal. The main refinement is that the final workbench uses the empirically selected G6 detector instead of assuming the autoencoder configuration must win.

B. Dataset and preprocessing

Q5. Why did you choose this dataset?

It is a recognised public benchmark with extreme class imbalance, real labelled fraud examples and sufficient scale for reproducible comparison. Its anonymity also makes it suitable for testing model-evidence fidelity, although it limits business interpretation.

Q6. Is the dataset large enough?

It is large enough for a controlled undergraduate experiment: 283,726 deduplicated transactions and 473 fraud cases. It is not sufficient for production generalisation because it is a single historical, anonymous and short-window dataset.

Q7. What happened during deduplication?

Content-based deduplication removed 1,081 repeated transaction rows. A stable case identifier was assigned before deduplication for traceability but excluded from the content comparison and from all model features.

Q8. Why use a 70/15/15 split?

Training data fits preprocessing and models, validation data controls early stopping, tuning and threshold selection, and test data provides the final locked evaluation. Stratification keeps the rare fraud class represented in each partition.

Q9. Why not use a chronological split?

A chronological split would better simulate temporal drift. The present stratified design supports stable minority-class comparison but does not prove future deployment performance. Temporal or rolling validation is future work.

Q10. How did you prevent leakage?

The split occurs before learned preprocessing. The scaler, SMOTE and autoencoder are fitted only on permitted training data. Validation controls tuning and threshold selection, the test set is used only after freezing, and case identifiers are excluded from the feature matrix.

Q11. Why standardise data if XGBoost uses trees?

XGBoost itself is usually insensitive to monotonic feature scaling, but the autoencoder and distance-based SMOTE are scale-sensitive. A consistent train-fitted representation also prevents preprocessing differences from confounding the experimental groups.

Q12. Does SMOTE create more real fraud data?

No. It creates synthetic training points between existing minority examples. It may help the learner, but it does not improve external validity or introduce new real fraud behaviours.

C. Detector design

Q13. Why XGBoost?

XGBoost is a strong and practical method for structured tabular data, handles nonlinear interactions and provides a suitable basis for controlled imbalance and feature-representation comparisons.

Q14. What does the autoencoder do?

It learns to reconstruct legitimate training transactions through a compressed representation. Reconstruction error and bottleneck activations are then evaluated as additional features for XGBoost. It is not the final fraud decision-maker.

Q15. Why train the autoencoder only on legitimate training cases?

The intended representation is the normal transaction manifold. Fraud and unusual legitimate cases may reconstruct less accurately, so reconstruction error becomes an anomaly-related feature.

Q16. Does high reconstruction error mean fraud?

No. It means unusual relative to the learned legitimate representation. A legitimate unusual case may also have high error, so it is used as a feature rather than a verdict.

Q17. Did Autoencoder-XGBoost win?

No clear and consistent advantage was observed. G2 and G7 were competitive, but the AP differences from the original-feature baselines were very small, while G3 was lower and more variable.

Q18. Then why keep Autoencoder in the project?

Because testing whether unsupervised anomaly and latent representations add value is a central research question. A well-supported negative result is useful and leads to the simpler operational choice.

Q19. Why did you choose G6 seed 42 for the app?

G6 had the numerically highest five-seed mean AP and strong precision. Seed 42 is the explicitly frozen run used to bind the detector, G4, G5 and workbench artifacts. The report does not claim G6 is statistically superior.

Q20. Why tune only G2 and G6?

They were the strongest untuned groups by validation AP and received a seeded 20-trial random search. This tuning asymmetry is transparently reported as a limitation because cross-group differences cannot be attributed solely to architecture or imbalance handling.

Q21. Why Average Precision?

It focuses on precision-recall performance for the rare positive class. Accuracy would be misleading, and ROC-AUC can look optimistic when the negative class is extremely large.

Q22. What is the meaning of precision 0.980392?

Of the 51 flagged test cases, 50 were fraud and one was legitimate. Precision describes queue purity, not the percentage of all frauds found.

Q23. What is the meaning of recall 0.704225?

The detector found 50 of the 71 fraud cases in the test split and missed 21.

Q24. Why is the threshold so high?

The value is on the model's score scale and was selected on validation data by maximising F1. It produced a high-precision review queue. It is not a universal banking policy or calibrated probability threshold.

Q25. Does a score of 1.0000 mean certainty?

No. The interface uses the score for ranking and thresholding. The model was not probability-calibrated, so the score must not be interpreted as exact real-world probability.

D. SHAP and explanations

Q26. What is G4?

G4 is the explanation stage. It applies Tree SHAP to the frozen G6 detector and stores the top three non-zero signed feature contributions for every flagged case as reason codes.

Q27. What exactly does SHAP explain?

It explains how input features contributed to the frozen model output relative to its baseline. It does not explain the real-world causal reason for fraud.

Q28. Why do many cases show V14?

The frozen model assigns V14 a large contribution for many flagged cases. The interface reflects the recorded evidence instead of artificially rotating features for visual variety.

Q29. Can you tell the analyst what V14 means?

No. V1 to V28 are anonymous PCA components. The system can say V14 pushed the model output toward fraud, but it cannot invent a merchant, device or behavioural meaning.

E. LLM and guardrails

Q30. What does the LLM receive?

It receives a bounded evidence package: a coarse risk bucket and ranked anonymous feature directions. It does not receive ground truth, raw transaction rows, exact feature values, detector probability or SHAP magnitude. The current live serializer also excludes case ID, amount and elapsed time.

Q31. Why use local Ollama?

Local inference reduces unnecessary external transmission and allows a pinned, reproducible runtime. It does not by itself prove security, privacy or regulatory compliance.

Q32. Why not use ChatGPT or a cloud API?

The project is designed around local data handling and offline reproducibility. A cloud API would add external disclosure, version and availability dependencies that are unnecessary for this FYP.

Q33. Why compare strict and simple prompts?

They preserve a comparable task while varying the strength of evidence constraints. The result measures prompt discipline rather than comparing unrelated output formats.

Q34. Why use the same raw output for OFF and ON policies?

It isolates the delivery policy. If I regenerated the narrative for each condition, generation randomness could be mistaken for a guardrail effect.

Q35. What does guardrail OFF mean?

It means measuring or hypothetically delivering the raw LLM output without validate-or-fallback enforcement. It does not turn off the detector or SHAP.

Q36. What does guardrail ON mean?

The same raw output is passed unchanged through the four deterministic checks. It is delivered only if all checks pass; otherwise the reason-code fallback is delivered.

Q37. What are the four checks?

Format, completeness, grounding and direction.

Q38. Why deterministic checks instead of another LLM judge?

Deterministic checks are repeatable, testable and auditable. Using an LLM to judge another LLM would add another generative uncertainty layer.

Q39. Does 330/330 mean the validator is perfect?

No. It means all attacks in the versioned, template-constrained calibration corpus were intercepted. Unknown failure forms may still pass, and valid unrestricted paraphrases may be rejected.

Q40. What is false rejection?

It is when a faithful narrative is rejected by the validator. Faithful controls are required because high attack detection alone could be achieved by rejecting everything.

Q41. Why was the simple prompt 51/51 invalid?

The simple prompt omitted detailed evidence-language constraints. Its output did not satisfy the strict closed delivery contract, especially format, grounding and direction. This is a result for one model, prompt design and case set, not a claim that all simple prompts always fail.

Q42. Why not silently normalise or repair the LLM output?

Repairing before measurement would hide the original model behaviour. The project stores the raw output unchanged, measures it, and either accepts it or replaces it with an authoritative fallback.

Q43. Why not always use the deterministic fallback?

That may be the safest operational default. The LLM is an optional research layer. This FYP measures whether generated wording can be safely bounded; it has not proven that the LLM is more useful than deterministic text.

Q44. Have you proven that analysts prefer the narrative?

No. The interim human evaluation used 11 proxy reviewers and 99 synthetic case reviews, so it supports descriptive preference and comprehension discussion only. It does not prove analyst productivity, professional analyst preference or decision-quality improvement.

Q45. Have you completed a human blind audit?

Yes. I completed a student-conducted blinded manual semantic audit of the 49 delivered strict ULB narratives. The result was 0/49 judged semantic violations, with a 95% Wilson interval of 0% to 7.27%. This supports the reviewed set, but it is a single-reviewer result and not a universal proof that all future outputs are safe.

F. Application and industrial realism

Q46. Is the application only a dashboard?

It is an analyst-oriented workflow prototype. It has a work queue, investigation workspace, narrative assurance, model/policy monitoring, persistent review state, activity history, evidence compatibility checks and provisional dispositions.

Q47. What can an analyst edit?

Only workflow metadata: status, provisional disposition and notes. Detector outputs, SHAP evidence, narratives, metrics and manifests remain read-only.

Q48. Where are analyst decisions stored?

In a separate local SQLite workflow database outside the experiment and report directories.

Q49. What happens if the evidence artifacts change?

The workflow record is compared with the current detector/G4/G5 evidence fingerprint. An incompatible old record is not silently reused as if it belonged to the new evidence chain.

Q50. What is optimistic concurrency?

Each workflow record has a revision. A stale browser must provide the expected revision and receives a conflict instead of overwriting a newer analyst update.

Q51. What happens if Ollama is down during the demo?

The live request returns a controlled unavailable state and deterministic reason-code fallback. The detector, SHAP evidence and recorded review workflow remain usable.

Q52. Is the application production ready?

No. It is a functional local research prototype with production-oriented boundaries and automated verification. It still lacks real banking integration, authentication, RBAC, enterprise security, drift operations and regulatory validation.

Q53. What tests currently pass?

On 20 July 2026, 37 backend dashboard tests, five React component tests and ten Playwright end-to-end tests passed. ESLint and the TypeScript/Vite production build also passed.

G. Novelty, value and limitations

Q54. What is your novelty in one line?

The novelty is the evaluated and fail-safe delivery boundary between frozen model evidence and generated analyst prose, not a new detector algorithm.

Q55. Is this only software engineering?

The software is the artefact, but the research lies in the controlled comparison and measurement: detector groups, prompt arms, paired delivery policies, adversarial calibration, false rejection, fallback and claim boundaries.

Q56. What is the social value?

It supports human investigators with consistent model evidence while reducing the risk that fluent but unsupported generated text is treated as fact. Local processing, auditability and graceful fallback are useful patterns for high-stakes decision support.

Q57. Could the system harm users?

Yes. False positives can inconvenience legitimate customers, false negatives can miss fraud, and fluent explanations can create automation bias. Human oversight, threshold governance, transparent evidence, fallback and monitoring are necessary mitigations.

Q58. What are the most important limitations?

One anonymous short-window dataset, a stratified rather than temporal split, small positive class, no external replication, one local LLM, 51 flagged ULB narratives, closed-grammar validator, single-reviewer semantic audit, proxy rather than professional human evaluation, and no production banking integration.

Q59. What would you do next with more time?

Add temporal and external validation, probability calibration, meaningful business features, independent multi-reviewer semantic-audit replication, professional analyst usability testing, a second local model, drift monitoring and enterprise workflow/security controls.

Q60. What is your final honest conclusion?

The detector comparison did not show a clear autoencoder advantage. The project nevertheless demonstrates a reproducible, auditable and fail-safe method for placing an optional local LLM behind frozen SHAP evidence and deterministic delivery controls in an analyst workflow.